Supplemental Material for all-particle energy spectrum and mean logarithmic mass of cosmic rays with LHAASO

For references see the main text.

Introduction.— The Tibet-III air shower array (4,300 m a.s.l.) estimates the cosmic ray primary energy based on the shower size fitted by a modified Nishimura-Kamata-Greisen (NKG) function [6]. The chemical composition and high-energy hadronic interaction models dependent systemic errors in flux were a few tens of percent in the energy range below the knee, 20% in the chemical composition models between heavy dominant (HD) and light dominant (protons and Helium, denoted by PD), and 10% in the high-energy hadronic interaction models between QGSJET01c and SIBYLL2.1. The IceTop experiment (2,835 m a.s.l.) estimated the cosmic ray energy using a reconstructed parameter (S_{125}) [8], with a relationship dependent on the primary mass and zenith angle. The IceCube/IceTop coincident analysis used neural network to reconstruct primary energy [9]. Their results were limited by the amount of data on hand, the systematic uncertainty due to detector effects (particularly the light yield in the ice), and the dependence on the choice of high-energy hadronic interaction model used for the simulations. The KASCADE experiment (110 m a.s.l.) determined the energy spectra for five primary mass groups by means of unfolding the two-dimensional frequency spectrum of electron and muon numbers [11]. None of the two interaction models can describe the data over the whole measurement range for the QGSJet01 and SIBYLL2.1 based analysis. The CASA-MIA experiment (870 g/cm^2) used a combination of parameters of "electron size" and muon size to reconstruct the cosmic ray energy [14, 15], even though the measurement isn't near shower maximum which is required by the Matthews-Heitler model [24].

The KM2A consists of 5,216 electromagnetic particle detectors (EDs) and 1,188 muon detectors (MDs), as shown in Fig. S1. The EDs and MDs can measure the electromagnetic particles and the muon content in air showers simultaneously with the primary energy from tens of TeV to tens of PeV. The MD of KM2A is the most powerful muon detector in current cosmic rays observatory on the ground, has shown its great ability in the measurement of Crab Nebula in the first phase operation of LHAASO [21]. EDs are distributed with a spacing of 15 m and MDs are distributed with a spacing of 30 m. More details about the experiment setup as described in Ref. [19].

The MD is a pure water Cherenkov detector with an inner diameter of 6.8 m and height of 1.2 m enclosed within a cylindrical concrete tank. An 8-inch PMT sits at the center of the top of the tank and looks downward. The thickness of overburden soil is 2.5 m to absorb the secondary electrons/positrons and gamma rays in showers[19]. This reduces the punch-through of electromagnetic particles [33]. The threshold for $\mu^{\pm}$ is about 1 GeV [22].

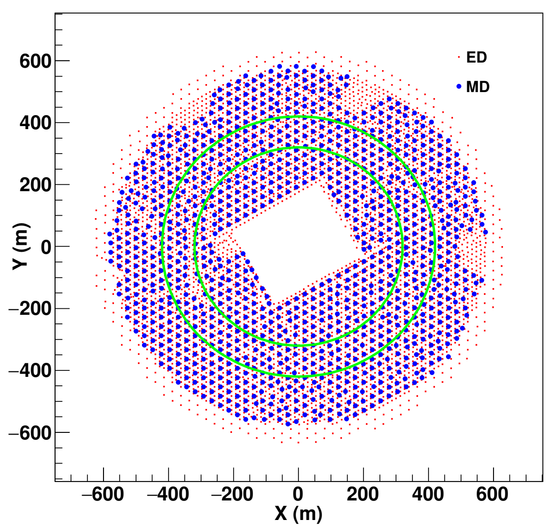

FIG. S1. The layout of LHAASO-KM2A detector. The red squares and blue dots indicate the EDs and MDs in operation, respectively. Showers are selected with reconstructed cores falling in the ring between the two green circles, inner and outer radii of 320 m and 420 m, respectively.

Data.—Air showers were simulated over a wide energy range, from 10^{13} to 5×10^{16} eV, using CORSIKA [27](v77410). The QGSJETII-04 [16], EPOS-LHC [17] and SIBYLL-2.3d [28] models for high-energy hadronic interactions were used, the FLUKA model [29] was adopted to simulate low-energy interactions, and the EGS4 model was used for electromagnetic interactions. For this study, five components − hydrogen (H), helium (He), nitrogen (denoted by CNO), aluminum (denoted by MgAlSi) and iron (Fe) − were generated. Further, showers following an E^{-2} energy spectrum and isotropic angular distribution were simulated within a zenith angle and azimuth angle ranges of $0°$-$40°$ and $0°$-$360°$, respectively.

Monte Carlo (MC) detector simulation were produced using a dedicated program developed by the collaboration based on the Geant4 package [30], named G4KM2A [31]. The core positions of the simulated samples fall within a circular ring region with a distance of 260-480 m from the center of the array, to avoid wasting simulated data. The total number of simulated events for the five components of the QGSJETII-04 model was approximately 5.555×10^7, and the number of simulated events per component is equal. The EPOS-LHC [17] simulation data was used for high-energy hadronic interaction model testing, and the simulated data samples were approximately 9.035×10^6. In addition, the SIBYLL-2.3d [28] simulated data samples were approximately 9.83×10^5 from 10^{14} to 10^{17} eV.

In order to reduce the pollution caused by the punch-through effect of high-energy electromagnetic particles near the shower core, the sum of muons counted by MD more than 40 m away from the shower core is denoted as N_μ [32]. Therefore, these muons, counting only the MDs within 40-200 m far from the shower axis, were used in the subsequent analysis. To have the same integral radius as muons, counting the electromagnetic particles N_e over EDs within 40–200 m from the shower axis was considered in later analysis. Both experimental and simulation events are processed through the same reconstruction routine. A detailed description of the detector simulation and the comparison between the experiment and simulation in a previous paper [21]. The simulated detector response agrees with the response of a real detector.

The data collected by LHAASO-KM2A array from September 2021 to December 2022 was selected. In order to study the performance of the detector, there are many artificial or unexpected factors that may cause data collection anomalies during operation. Hence, for this analysis, only data obtained during normal operation were utilized. The total live time (T) for the analysis is approximately 375 days.

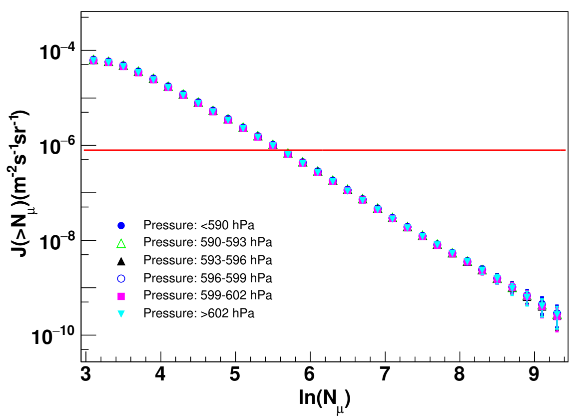

FIG. S2. Muon integral intensities for different pressure intervals derived from the measurements with LHAASO-KM2A. The CIC employed is shown as horizontal red line ($\log_{10}(J/(\mathrm{m}^2 \cdot \mathrm{s} \cdot \mathrm{sr})) = -6.1$). Error bars represent statistical uncertainties.

A maximum variation of about 12 hPa is observed in the atmospheric pressure. Here, we will use the approach based on the Constant Intensity Cut (CIC) method [25, 26], to correct the effect of atmospheric pressure on the experimental data. The aim of the CIC method is to provide a way to relate data from different atmospheric pressure at roughly the same primary energies, without any reference to MC simulations. In order to apply the CIC method, in the first instance, data were grouped into six pressure intervals as shown in Fig. S2, and their intervals are [<590] hPa, [590-593] hPa, [593-596] hPa, [596-599] hPa, [599-602] hPa, and [>602] hPa. The integral muon intensity $J(> N_\mu)/(\mathrm{m}^2 \cdot \mathrm{s} \cdot \mathrm{sr})$ is estimated according to the differential muon shower size spectrum. After cutting according to $\log_{10}(J/(\mathrm{m}^2 \cdot \mathrm{s} \cdot \mathrm{sr}) = -6.1$ in Fig. S2, the number of muons under different air pressure is obtained, and the red dots are shown in Fig. S3. The measured muon contents vary with the atmospheric pressure and can be described by quadratic functions, $N_\mu = a_0 + a_1 \cdot P + a_2 \cdot P^2$, here P represents atmospheric pressure. To be able to study atmospheric changes and apply corrections where needed, the ground pressure at an altitude of 4,397 m is 586.6 hPa (CORSIKA atmosphere 1 [27]) as the reference atmosphere. The influence of atmospheric pressure on the measurement of muon content was 4.4%, and become 0.6% after the correction of atmospheric pressure, as shown in Fig. S3. Of course, there is a slight difference within a few thousandths at different intensity $\log_{10}(J/(\mathrm{m}^2 \cdot \mathrm{s} \cdot \mathrm{sr})$ cuts, the modified air

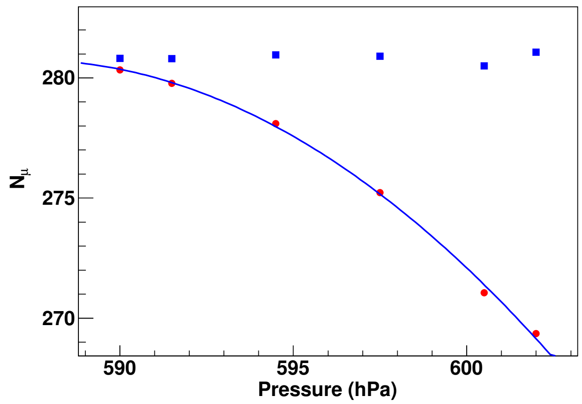

FIG. S3. After the CIC method, the measured muon contents vary with atmospheric pressure, and the blue line is fitted with quadratic law functions. The red dots are the measured values, and the blue squares are the muon content corrected for atmospheric pressure.

pressure data affects the measurements of all-particle energy spectrum within ±3% and the mean logarithmic mass of cosmic rays within ±4%.

Event selection.— To maintain good data quality, the events are selected with the following criteria. 1) To capture cosmic ray showers near their maximum within a range of two energy decades, zenith angles ranging from 10° (610 g/cm^2) to 30° (690 g/cm^2) are selected; 2) The location of the reconstructed shower core was restricted to inner and outer ring radii of 320 and 420 m, respectively. The core of the cosmic ray shower is located far from the edges of the KM2A array, both the inner and outer edges, ensuring that muons or electromagnetic particles are not lost within the range of 40-200 m. This selection helps to minimize the possibility of erroneously reconstructing showers with cores outside the array. Consequently, events are selected with the shower core positioned within the green ring (320-420 m) as illustrated in Fig. S1. Thus, an erroneous reconstruction of showers with cores outside the array can be avoided, thereby reducing the impact of edge effects. The results of the simulation data show that this shower core selection is sufficient for simulation data of sampling in the circular ring region of 260–480 m; 3) The number of particles N_μ, was larger than 0 while N_e was larger than 80. After the above data quality selections, the selected samples reached full efficiency when the energy exceeded 300 TeV. Therefore, the following data analysis was based on the samples after achieving full efficiency. Figure S4 shows the effective aperture of five mass groups after the data quality selections. Above 300 TeV, all the five mass groups of cosmic ray components reach full detection efficiency with an calculated aperture of 0.16 km^{2}sr.

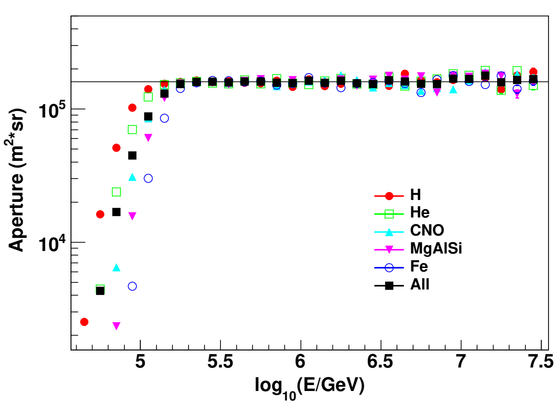

FIG. S4. The aperture of the KM2A full-array for cosmic ray showers vs. the primary energy of each mass group. The black line is the calculated aperture 0.16 km^{2}sr. The error bars show the statistical uncertainties.

Energy reconstruction.—The energy reconstruction formula used is $\log_{10}(E/\mathrm{GeV}) = p_0 + p_1 \cdot \log_{10}(N_{\mathrm{e}\mu})$. The resolution and bias of the reconstructed energy are shown in Fig. S5, where the resolution is the fitted Gaussian sigma of the $(E_\mathrm{rec}$-$E_\mathrm{true})/E_\mathrm{true}$ distribution, while the bias is the mean value of this distribution for all components together. The bias is less than 6% for different individual components. The resolution is better than 15% above PeV, and decreases further as the energy increases.

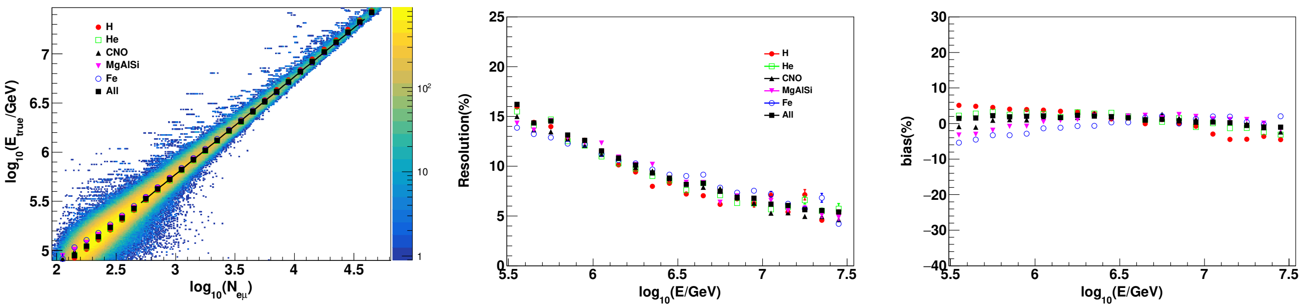

FIG. S5. Left:Using the QGSJETII-04 model as an example, the correlation between the true energy and the parameter $N_{e\mu}$ of the air shower is based on simulation data from different cosmic ray components. "All" represents the weighted values of all components according to the Gaisser H3a spectrum[34]. The color scale on the plot indicates the number of simulated events after the selection process. Middle: the reconstructed energy resolution for individual and all components; Right: the reconstructed energy bias for individual and all components.

Using the QGSJETII-04 model as an example, the fit parameters p_0 and p_1 for Gaisser H3a [34], Horandel [35], GST [36] and GSF [37] composition models can be found in Table S1. So the average one is used to reconstruct the cosmic ray energy, which is $\log_{10}(E/\mathrm{GeV}) = 2.799 + 0.992 \cdot \log_{10}(N_{e\mu})$ for QGSJETII-04 model.

TABLE S1. The fit parameters p_0 and p_1 for different composition models.

Model	p_0	p_1
Gaisser	2.799	0.992
Horandel	2.798	0.992
GST	2.802	0.992
GSF	2.797	0.992

The fit parameters p_0 and p_1 for the QGSJETII-04, EPOS-LHC and SIBYLL-2.3d high-energy hadronic interaction models can be found in Table S2.

TABLE S2. The fit parameters p_0 and p_1 for different high-energy hadronic interaction models.

Model	p_0	p_1
QGSJETII-04	2.799	0.992
EPOS-LHC	2.789	0.992
SIBYLL-2.3d	2.784	0.995

All-particle energy spectrum of cosmic rays.—In order to verify the correctness of the all-particle energy spectrum measurement method, we use simulated data to check the measurement method. For example, we use the Gaisser H3a model [34], after the reconstruction of energy, the points in the Fig. S6 represent the energy spectrum obtained from the analysis of simulation data, and the lines are the expected Gaisser H3a energy spectrum. The validation of Equation (3) is a full end-to-end validation of the whole analysis chain. The energy spectrum obtained from the results is consistent with the expected energy spectrum model.

For QGSJETII-04 model, the knee position is a value of 3.72$\pm$0.05 PeV, and the spectral indices are -2.7430 ± 0.0004 and -3.131 ± 0.005 before and after the knee, respectively. The sharpness of the knee is a value of 4.1 ± 0.1. This study represents a precise measurement of the all-particle energy spectrum to date. The EPOS-LHC and SIBYLL-2.3d models are employed as alternative high-energy hadronic interaction model, and the resulting all-particle energy spectrum is also presented in Fig. S9. For the EPOS-LHC model, the knee position is determined to be 3.62$\pm$0.05 PeV, while the spectral indices are -2.7430 ± 0.0004 and -3.130 ± 0.005 before and after the knee, respectively. The

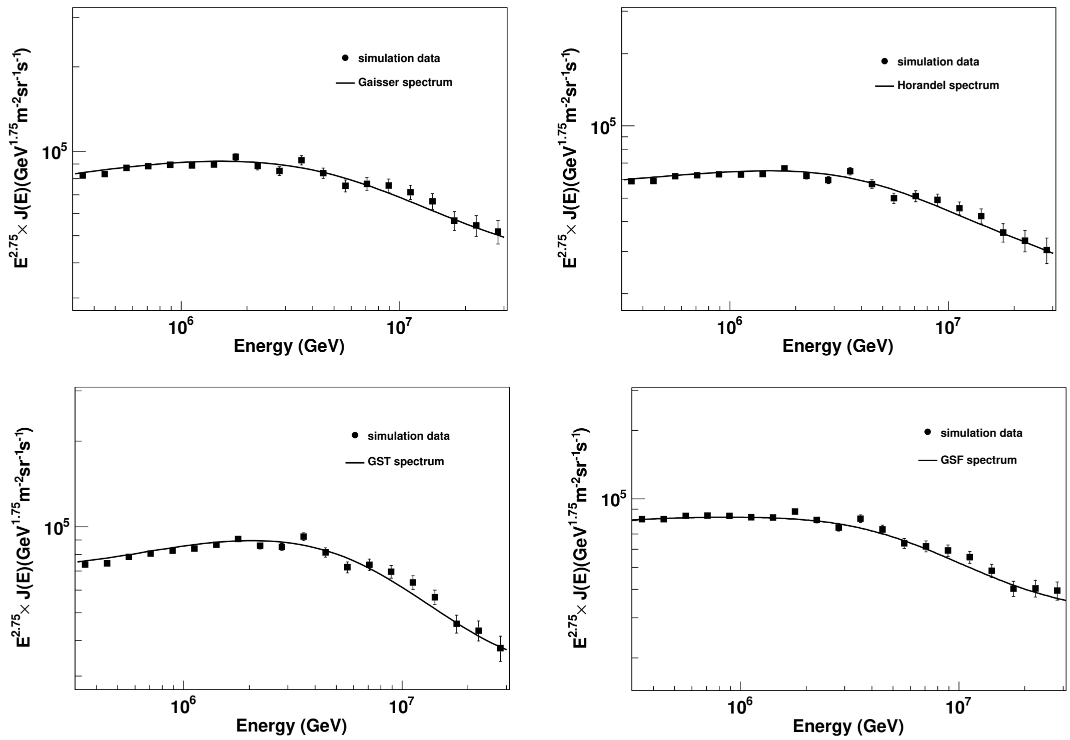

FIG. S6. The assumption cosmic rays spectrum as solid curve indicated in plot, Gaisser H3a [34], Horandel [35], GST [36] and GSF [37] following the truth energy, and the dots represent the energy spectrum obtained from the analysis of simulation data following the reconstructed energy.

TABLE S3. List of systematic errors (percent error) on flux of all-particle energy spectrum and the mean logarithmic mass of cosmic rays.

	Flux $J(E)$	$\langle \ln(A) \rangle$
Air preesure	$\pm 3\%$	$\pm 4\%$
Composition models	$\pm 1.5\%$	$\pm 3\%$
Interaction models	$\pm 2.5\%$	$\pm 6\%$

sharpness of the knee is quantified as 4.2 ± 0.1. For the SIBYLL-2.3d model, the knee position is determined to be 3.67 ± 0.04 PeV, while the spectral indices are -2.7377 ± 0.0004 and -3.112 ± 0.004 before and after the knee, respectively. The sharpness of the knee is quantified as 3.7 ± 0.1. The fit results for different high-energy hadronic interaction models can be found in Table S4.

The values of the all-particle energy spectrum from the KM2A full-array analysis, the corresponding statistical and systematic uncertainties, are listed in Table S5.

Mean logarithmic mass of cosmic rays.—The parameters x_0 and x_1 can be obtained by fitting the plot as shown in Fig. S7 as examples for one energy interval. The $\langle \ln(N_\mu) \rangle$ is the mean of the logarithmic of muon content for the

TABLE S4. The fit results for different high-energy hadronic interaction models of all-particle energy spectrum (Equation (4))

Model	E_b (PeV)	γ_1	γ_2	s
QGSJETII-04	3.72 ± 0.05	-2.7431 ± 0.0004	-3.131 ± 0.005	4.1 ± 0.1
EPOS-LHC	3.62 ± 0.05	-2.7431 ± 0.0004	-3.130 ± 0.005	4.2 ± 0.1
SIBYLL-2.3d	3.67 ± 0.04	-2.7377 ± 0.0004	-3.112 ± 0.004	3.7 ± 0.1

TABLE S5. All-particle energy spectrum flux.

Energy $\log_{10}(E/\mathrm{GeV})$	flux±stat. ±syst.[a](QGSJETII-04) $\mathrm{GeV}^{-1}\mathrm{m}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$	flux±stat.±syst. (EPOS-LHC) $\mathrm{GeV}^{-1}\mathrm{m}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$	flux±stat. ±syst.(SIBYLL-2.3d) $\mathrm{GeV}^{-1}\mathrm{m}^{-2}\mathrm{s}^{-1}\mathrm{sr}^{-1}$
5.5-5.6	$(5.3527 \pm 0.0011 \pm 0.1795) \times 10^{-11}$	$(5.1413 \pm 0.0011 \pm 0.1724) \times 10^{-11}$	$(5.1952 \pm 0.0011 \pm 0.1743) \times 10^{-11}$
5.6-5.7	$(2.8462 \pm 0.0007 \pm 0.0955) \times 10^{-11}$	$(2.7351 \pm 0.0007 \pm 0.0917) \times 10^{-11}$	$(2.7665 \pm 0.0007 \pm 0.0928) \times 10^{-11}$
5.7-5.8	$(1.5163 \pm 0.0005 \pm 0.0509) \times 10^{-11}$	$(1.4567 \pm 0.0005 \pm 0.0489) \times 10^{-11}$	$(1.4754 \pm 0.0005 \pm 0.0495) \times 10^{-11}$
5.8-5.9	$(8.0536 \pm 0.0031 \pm 0.2701) \times 10^{-12}$	$(7.7363 \pm 0.0030 \pm 0.2595) \times 10^{-12}$	$(7.8437 \pm 0.0030 \pm 0.2631) \times 10^{-12}$
5.9-6.0	$(4.2741 \pm 0.0020 \pm 0.1434) \times 10^{-12}$	$(4.1049 \pm 0.0020 \pm 0.1377) \times 10^{-12}$	$(4.1687 \pm 0.0020 \pm 0.1398) \times 10^{-12}$
6.0-6.1	$(2.2710 \pm 0.0013 \pm 0.0762) \times 10^{-12}$	$(2.1816 \pm 0.0013 \pm 0.0732) \times 10^{-12}$	$(2.2175 \pm 0.0013 \pm 0.0744) \times 10^{-12}$
6.1-6.2	$(1.2071 \pm 0.0008 \pm 0.0405) \times 10^{-12}$	$(1.1597 \pm 0.0008 \pm 0.0390) \times 10^{-12}$	$(1.1804 \pm 0.0008 \pm 0.0396) \times 10^{-12}$
6.2-6.3	$(6.4186 \pm 0.0055 \pm 0.2153) \times 10^{-13}$	$(6.1620 \pm 0.0054 \pm 0.2067) \times 10^{-13}$	$(6.2818 \pm 0.0054 \pm 0.2107) \times 10^{-13}$
6.3-6.4	$(3.3875 \pm 0.0035 \pm 0.1136) \times 10^{-13}$	$(3.2514 \pm 0.0035 \pm 0.1091) \times 10^{-13}$	$(3.3186 \pm 0.0035 \pm 0.1113) \times 10^{-13}$
6.4-6.5	$(1.7715 \pm 0.0023 \pm 0.0594) \times 10^{-13}$	$(1.6972 \pm 0.0022 \pm 0.0569) \times 10^{-13}$	$(1.7366 \pm 0.0023 \pm 0.0582) \times 10^{-13}$
6.5-6.6	$(9.1282 \pm 0.0146 \pm 0.3062) \times 10^{-14}$	$(8.7356 \pm 0.0143 \pm 0.2930) \times 10^{-14}$	$(8.9576 \pm 0.0145 \pm 0.3004) \times 10^{-14}$
6.6-6.7	$(4.6234 \pm 0.0092 \pm 0.1551) \times 10^{-14}$	$(4.4207 \pm 0.0091 \pm 0.1483) \times 10^{-14}$	$(4.5391 \pm 0.0092 \pm 0.1522) \times 10^{-14}$
6.7-6.8	$(2.3071 \pm 0.0058 \pm 0.0774) \times 10^{-14}$	$(2.2003 \pm 0.0057 \pm 0.0738) \times 10^{-14}$	$(2.2680 \pm 0.0058 \pm 0.0761) \times 10^{-14}$
6.8-6.9	$(1.1314 \pm 0.0036 \pm 0.0379) \times 10^{-14}$	$(1.0760 \pm 0.0036 \pm 0.0361) \times 10^{-14}$	$(1.1123 \pm 0.0036 \pm 0.0373) \times 10^{-14}$
6.9-7.0	$(5.4259 \pm 0.0225 \pm 0.1820) \times 10^{-15}$	$(5.1382 \pm 0.0219 \pm 0.1723) \times 10^{-15}$	$(5.3367 \pm 0.0223 \pm 0.1790) \times 10^{-15}$
7.0-7.1	$(2.6484 \pm 0.0140 \pm 0.0888) \times 10^{-15}$	$(2.5301 \pm 0.0137 \pm 0.0849) \times 10^{-15}$	$(2.6140 \pm 0.0139 \pm 0.0877) \times 10^{-15}$
7.1-7.2	$(1.2835 \pm 0.0087 \pm 0.0430) \times 10^{-15}$	$(1.2240 \pm 0.0085 \pm 0.0411) \times 10^{-15}$	$(1.2697 \pm 0.0087 \pm 0.0426) \times 10^{-15}$
7.2-7.3	$(6.3841 \pm 0.0547 \pm 0.2141) \times 10^{-16}$	$(6.0622 \pm 0.0533 \pm 0.2033) \times 10^{-16}$	$(6.3133 \pm 0.0544 \pm 0.2118) \times 10^{-16}$
7.3-7.4	$(3.1220 \pm 0.0341 \pm 0.1047) \times 10^{-16}$	$(3.0000 \pm 0.0334 \pm 0.1006) \times 10^{-16}$	$(3.0989 \pm 0.0340 \pm 0.1039) \times 10^{-16}$
7.4-7.5	$(1.5795 \pm 0.0216 \pm 0.0530) \times 10^{-16}$	$(1.5112 \pm 0.0211 \pm 0.0507) \times 10^{-16}$	$(1.5692 \pm 0.0215 \pm 0.0526) \times 10^{-16}$

[a] syst.: excluding the uncertainty from hadronic interaction models.

cosmic ray with mass A within the energy interval. These parameters change slowly as energy within 0.2 intervals between $\log_{10}(E_{\rm rec}/\mathrm{GeV})$ =5.5 and 7.5. Since the parameters x_0 and x_1 are obtained through simulation data fitting, in order to reduce the statistical fluctuation of simulation data, the energy interval of simulation data is 0.2. By fitting the relation between x_0 or x_1 and cosmic rays energy, x_0 or x_1 value within 0.1 energy interval can be obtained by interpolation.

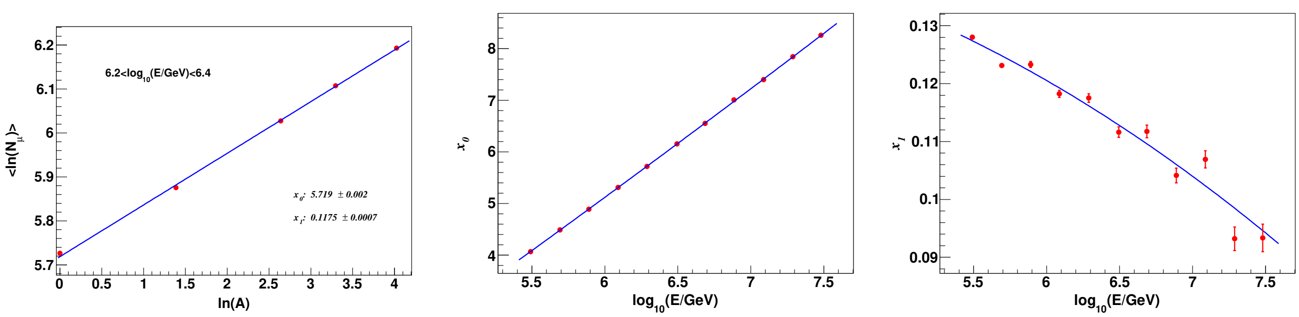

FIG. S7. Left:Using the QGSJETII-04 model as an example to demonstrate the relationship between $\langle \ln(N_\mu)\rangle$ and $\langle \ln(A)\rangle$ within a specific energy interval. The data points represent different mass groups, including H, He, CNO, MgAlSi, and Fe. The error bars represent the statistical uncertainties. The blue solid curve represents the fit of Equation (6) to the data. Middle and Right: The x_0 and x_1 as a function of energy, and fitted by quadratic functions. The error bars show the statistical uncertainties.

The real mean logarithmic mass of Gaisser H3a model [34] following the truth energy is shown as the black curve in Fig. S8. With the Equation (6), the mean logarithmic mass ($\langle \ln(A)\rangle$) of the simulation events as derived from $\langle \ln(N_\mu)\rangle$ within the reconstructed energy is also plotted and this derived mass almost overlaps with the assuming composition model. These plots verify that the $\langle \ln(A)\rangle$ can reliably be derived from the $\langle \ln(N_\mu)\rangle$ following Equation (6) within uncertainties.

With above method, the $\langle \ln(A)\rangle$ is measured by LHAASO in each energy bin from 0.3 PeV to 30 PeV and the results from three high-energy hadronic interaction models (QGSJETII-04, EPOS-LHC, and SIBYLL-2.3d models) are shown in Fig. S9. The values of $\langle \ln(A)\rangle$, along with their corresponding statistical and combined systematic uncertainties (excluding the uncertainty from hadronic interaction models), are included in Table S6. The fit results for EPOS-LHC, QGSJETII-04 and SIBYLL-2.3d high-energy hadronic interaction models of $\langle \ln(A) \rangle$ can be found in Table S7.

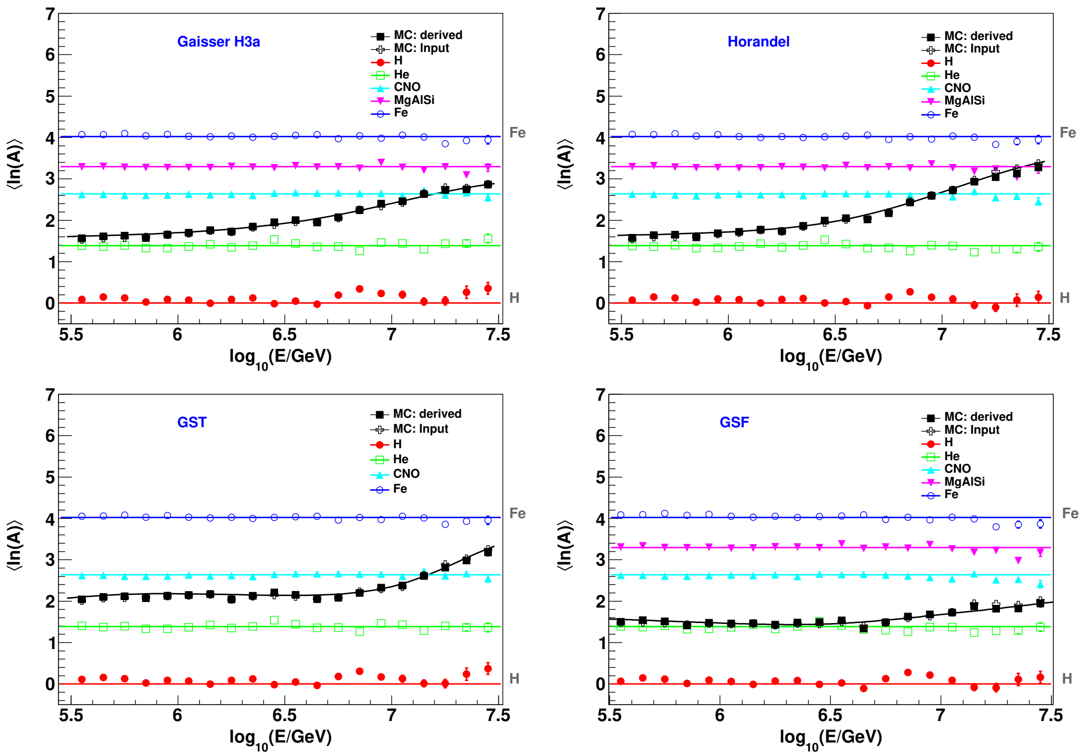

FIG. S8. The plots of measured $\langle \ln(A) \rangle$ as a function of the reconstructed energy. The black square represents $\langle \ln(A) \rangle$ derived from $\langle \ln(N_\mu) \rangle$ of the assumption energy spectrum, and five colors styles represent individual components $\langle \ln(A) \rangle$ derived from $\langle \ln(N_\mu) \rangle$, respectively. The black open-cross (MC:input) is the truth $\langle \ln(A) \rangle$ of the simulated events within the reconstructed energy bin. The black curve is the counting result based on the spectrum according to Gaisser H3a [34], Horandel [35], GST [36] and GSF [37] models following the truth energy. The error bars show the statistical uncertainties.

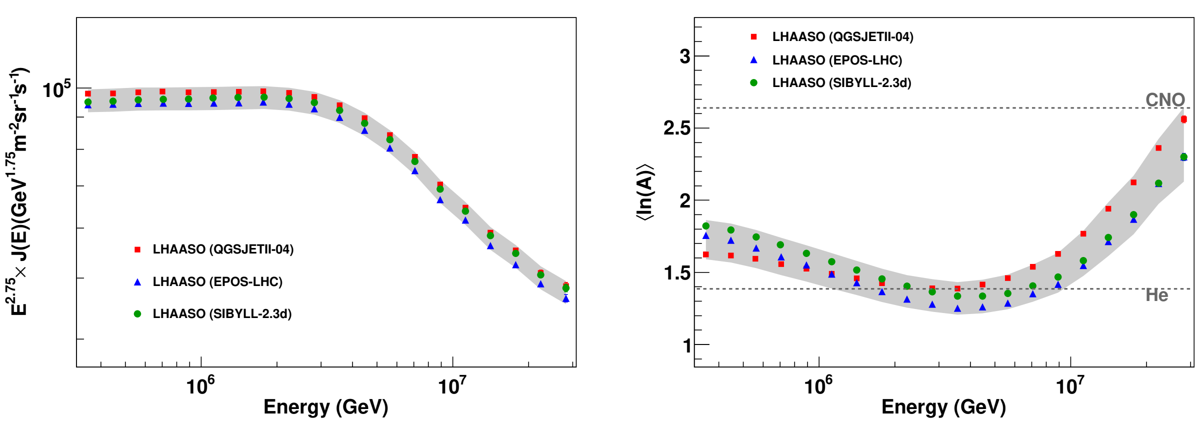

FIG. S9. Left: The all-particle energy spectrum flux multiplied by $E^{2.75}$ as a function of energy; Right: The $\langle \ln(A) \rangle$ of cosmic rays as a function of energy. The QGSJETII-04, EPOS-LHC and SIBYLL-2.3d hadronic interaction models simulated dataset are used to reconstruct these spectrum as the dots in different colors respectively. The error bars show the statistical uncertainties and the gray shadow band represents the estimated combined systematic uncertainties.

TABLE S6. The mean logarithmic mass of cosmic rays.

Energy $\log_{10}(E/\mathrm{GeV})$	$\langle\ln(A)\rangle$ ±stat. ±syst.[a] (QGSJETII-04)	$\langle\ln(A)\rangle$ ±stat.±syst. (EPOS-LHC)	$\langle\ln(A)\rangle$ ±stat. ±syst. (SIBYLL-2.3d)
5.5-5.6	$1.6246 \pm 0.0004 \pm 0.0819$	$1.7562 \pm 0.0005 \pm 0.0915$	$1.8221 \pm 0.0005 \pm 0.0950$
5.6-5.7	$1.6170 \pm 0.0005 \pm 0.0815$	$1.7224 \pm 0.0006 \pm 0.0915$	$1.7930 \pm 0.0006 \pm 0.0953$
5.7-5.8	$1.5907 \pm 0.0006 \pm 0.0802$	$1.6697 \pm 0.0007 \pm 0.0909$	$1.7450 \pm 0.0007 \pm 0.0951$
5.8-5.9	$1.5563 \pm 0.0008 \pm 0.0785$	$1.6083 \pm 0.0008 \pm 0.0903$	$1.6875 \pm 0.0008 \pm 0.0947$
5.9-6.0	$1.5250 \pm 0.0009 \pm 0.0771$	$1.5507 \pm 0.0010 \pm 0.0903$	$1.6325 \pm 0.0010 \pm 0.0950$
6.0-6.1	$1.4905 \pm 0.0011 \pm 0.0756$	$1.4881 \pm 0.0012 \pm 0.0905$	$1.5734 \pm 0.0012 \pm 0.0955$
6.1-6.2	$1.4579 \pm 0.0013 \pm 0.0742$	$1.4276 \pm 0.0015 \pm 0.0913$	$1.5155 \pm 0.0014 \pm 0.0965$
6.2-6.3	$1.4246 \pm 0.0016 \pm 0.0729$	$1.3674 \pm 0.0018 \pm 0.0925$	$1.4551 \pm 0.0017 \pm 0.0979$
6.3-6.4	$1.4010 \pm 0.0020 \pm 0.0722$	$1.3155 \pm 0.0022 \pm 0.0946$	$1.4040 \pm 0.0021 \pm 0.1001$
6.4-6.5	$1.3901 \pm 0.0024 \pm 0.0722$	$1.2803 \pm 0.0027 \pm 0.0978$	$1.3654 \pm 0.0026 \pm 0.1034$
6.5-6.6	$1.3867 \pm 0.0030 \pm 0.0729$	$1.2511 \pm 0.0033 \pm 0.1016$	$1.3346 \pm 0.0032 \pm 0.1073$
6.6-6.7	$1.4154 \pm 0.0037 \pm 0.0753$	$1.2605 \pm 0.0041 \pm 0.1072$	$1.3356 \pm 0.0040 \pm 0.1130$
6.7-6.8	$1.4608 \pm 0.0046 \pm 0.0788$	$1.2866 \pm 0.0051 \pm 0.1138$	$1.3545 \pm 0.0050 \pm 0.1196$
6.8-6.9	$1.5386 \pm 0.0057 \pm 0.0841$	$1.3512 \pm 0.0064 \pm 0.1222$	$1.4058 \pm 0.0062 \pm 0.1279$
6.9-7.0	$1.6279 \pm 0.0073 \pm 0.0903$	$1.4172 \pm 0.0083 \pm 0.1309$	$1.4670 \pm 0.0080 \pm 0.1369$
7.0-7.1	$1.7683 \pm 0.0092 \pm 0.0994$	$1.5483 \pm 0.0104 \pm 0.1427$	$1.5816 \pm 0.0101 \pm 0.1487$
7.1-7.2	$1.9414 \pm 0.0118 \pm 0.1105$	$1.7135 \pm 0.0132 \pm 0.1565$	$1.7412 \pm 0.0129 \pm 0.1628$
7.2-7.3	$2.1233 \pm 0.0147 \pm 0.1226$	$1.8675 \pm 0.0165 \pm 0.1706$	$1.8992 \pm 0.0162 \pm 0.1775$
7.3-7.4	$2.3612 \pm 0.0184 \pm 0.1383$	$2.1163 \pm 0.0203 \pm 0.1901$	$2.1185 \pm 0.0204 \pm 0.1967$
7.4-7.5	$2.5605 \pm 0.0225 \pm 0.1530$	$2.2997 \pm 0.0251 \pm 0.2075$	$2.3011 \pm 0.0251 \pm 0.2148$

[a] syst.: excluding the uncertainty from hadronic interaction models.

TABLE S7. The fit results for different high-energy hadronic interaction models of $\langle\ln(A)\rangle$ (Equation (4))

Model	E_b[a] (PeV)	γ_1	γ_2	s
QGSJETII-04	5.00 ± 0.07	-0.0777 ± 0.0003	0.368 ± 0.006	5.2 ± 0.2
EPOS-LHC	5.34 ± 0.07	-0.1483 ± 0.0003	0.386 ± 0.006	6.2 ± 0.3
SIBYLL-2.3d	5.82 ± 0.08	-0.1330 ± 0.0003	0.367 ± 0.008	6.3 ± 0.3

[a] E_b: The energy where the composition of cosmic rays shows a clear change from Helium on average toward the heavier CNO on average.